CloudTrie: An Uncertainty-Quantified Anomaly Detection Framework for Prefix Hijacking With Multisource Fusion

Zheng Wu, Shiyu Liu, Ruixi Xie, Ming Gao, *Member, IEEE*, Zhan Wu, Lei Han, Xiaobai Chen, *Member, IEEE*, and Fu Xiao, *Senior Member, IEEE*

***Abstract*—Anomaly detection is a crucial function for Border Gateway Protocol (BGP) security, providing the ability to trigger alarms and locate where the anomalies occur. Most recently, BGP prefix hijacking incidents have proliferated significantly. However, the existing detection methods exhibit limited accuracy, primarily because the BGP routing data they operate on is inherently uncertain. To address this challenge, we propose *CloudTrie*, an uncertainty-aware framework for online prefix hijacking detection. First, we quantify data uncertainty through a comprehensive measurement study of three BGP-related data sources (BGP RIB, RPKI, and IRR), assessing incompleteness, temporal variability, and multiple origin AS (MOAS) conflicts. Building on this, we leverage cloud modeling theory to construct spatio-temporal prefix/origin AS (P/O) cloud drops for dynamic uncertainty scoring of P/O pairs. Finally, we propose an online Trie-based framework to achieve instantaneous prefix hijack identification and online update. Experiments demonstrate that *CloudTrie* processes 117 thousand route updates per second with low memory overhead (34.3MB) and detects all of the anomaly events with less false alarms than the state-of-the-art. These results validate *CloudTrie* as a robust foundation for accurate, real-time BGP anomaly detection (AD).**

***Index Terms*—Border Gateway Protocol (BGP), cloud model theory, prefix hijacking, Trie, uncertainty measurement.**

I. Introduction

ANOMALY detection (AD) is an imperative function for prolonged monitoring of complex systems containing large volumes of data. In most complex scenarios, the obtained data generally contains amounts of noise, leading to uncertainty in these data. This data uncertainty challenge is particularly pronounced in the domain of Border Gateway Protocol (BGP) AD. The BGP is the core routing protocol of the Internet, responsible for exchanging routing information on reachability between autonomous systems (ASes). However, BGP was not designed with security mechanisms from its inception [1], leading to frequent anomalies that are characterized by rapid propagation and wide-ranging impact. In recent decades, as Internet infrastructure has become increasingly complex and network scale continues to expand, the occurrence frequency of inter-domain routing anomalies has been significantly rising [2], leading to significant negative consequences for the global network, even causing severe economic losses and social panic [3]. Among BGP anomalies, prefix hijacking accounts for the most significant proportion, at approximately 96.48%.[1] Therefore, the defense of prefix hijacking has become a critical issue for ensuring the secure and stable operation of the Internet's infrastructure.

In this context, Internet service providers (ISPs) generally deploy the monitoring instrument for detecting BGP hijacking anomalies in time [4]. The AD methods leverage passively or actively acquired routing data to monitor abnormal behaviors. Once the anomaly occurs and is detected, network operators will take measures to mitigate the negative impact of anomalies. Thus, to enable operators to respond promptly, the most advanced instruments for BGP AD need to accurately monitor BGP behaviors in real time.

The existing AD methods can be classified into information repository-based, active probing-based, and machine learning-based approaches. The active probing-based methods detect connectivity between ASes through active probing instruments like Ping and Tracert commands [5]. However, as the BGP topology becomes increasingly complex and large, the overhead of these methods grows substantially, making real-time, comprehensive monitoring of the BGP network infeasible. Machine learning-based methods perform detection by extracting temporal or spatial features to analyze patterns of abnormal behavior [6]. Although these methods possess good generalization performance and inference capabilities, they rely on large volumes of accurate data labels and suffer from poor interpretability [7]. Information repository-based AD methods depend on constructing a complete and accurate mapping between prefixes and their origin ASes, and then performing

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[1] https://bgpwatch.cgtf.net/#/

detection via a lookup-and-match process. This type of method offers strong real-time performance and facilitates root-cause analysis [8]. Nevertheless, such methods heavily depend on the accuracy of the input data; otherwise, the reliability of detection cannot be ensured [9].

Unfortunately, obtaining accurate and complete data in the large-scale BGP network remains a significant challenge due to several inherent sources of uncertainty. First, owing to the distributed nature of the infrastructure and the partial view provided by BGP collectors, the collected routing data is often inadequate for comprehensive measurement. Furthermore, the lack of a robust route authentication renders BGP messages susceptible to erroneous or malicious announcements [10]. Finally, BGP data is inherently dynamic and time-varying; routing information can quickly become stale, reducing the reliability of historical data for current detection tasks. These factors collectively introduce substantial uncertainty into the observed data, which in turn severely degrades the accuracy and robustness of existing AD methods [11].

Therefore, a core challenge in achieving accurate prefix hijacking detection lies in extracting sufficiently accurate data from vast, uncertain BGP routing data. It is crucial to build an AD system that is robust to uncertain data and capable of providing explainable results. To address this challenge, we propose *CloudTrie*, an uncertainty-aware AD instrumentation, fusing multiple data sources designed to identify prefix hijacking under the uncertain BGP data robustly.

The contributions are highlighted as follows.

1) *Comprehensive uncertainty measurement on BGP-related data:* To assess the extent of uncertainty in BGP data, we conduct a comprehensive measurement study on three key BGP-related routing datasets, focusing on three primary sources of uncertainty: data incompleteness, temporal variability, and the multiorigin AS (MOAS) phenomenon. Our results reveal that each dataset exhibits distinct characteristics and varying degrees of uncertainty, highlighting the complexity and challenges in achieving reliable BGP monitoring and analysis.
2) *Real-time uncertainty assessment for BGP routes:* To evaluate each P/O pair, we design an online route credibility assessment method based on a cloud model. First, this method extracts knowledge considering 3-D, i.e., time, space, and data source, based on a constructed observation matrix. Then, we structure a cloud model to assess and filter each route in real-time to convert uncertain routing data into data quantified with a degree of uncertainty.
3) *Temporal-adaptive detection framework:* To address the temporal variability of BGP routing data, we propose an update framework. Specifically, we construct two Tries, namely, ModelTrie and DetectTrie, responsible for model construction and online detection, respectively. Additionally, we apply two dedicated update mechanisms, i.e., offline and online mechanism so as to overcome the time-varying nature of BGP routing data.

Our *CloudTrie* system includes comprehensive instrumentation designed to accurately and effectively detect BGP prefix hijacking anomalies. We conduct extensive experiments on two real-world datasets, namely, closed and open datasets, to evaluate *CloudTrie*'s performance in terms of robustness, efficiency, and accuracy. Compared to state-of-the-art baselines, *CloudTrie* achieves an average throughput of 117 thousand route update messages per second while maintaining low memory consumption (under 34.3MB). Crucially, it detects all of the anomaly events with less false alarms, outperforming the state-of-the-art methods. *CloudTrie*'s source code and the labeled dataset are publicly available at https://github.com/185399144/cloudtrie.git

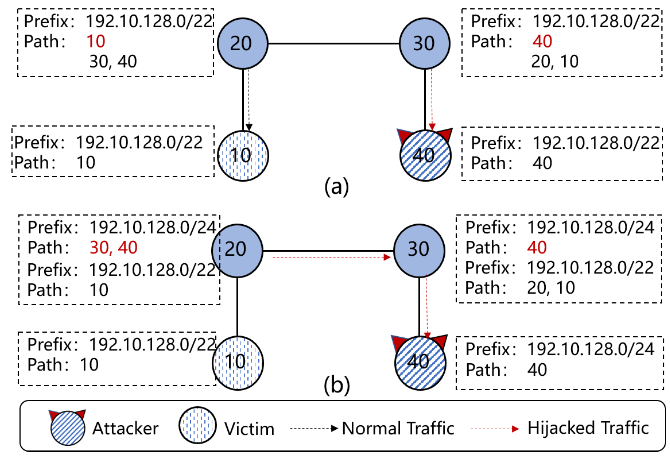

Fig. 1. Attack models of BGP hijacking. (a) Prefix hijacking. (b) Sub-prefix hijacking.

The remaining sections of this article are organized as follows. Section II introduces the preliminary and the problem we research. Section III measures the uncertainty in BGP-related data. Section IV presents the design of our detection system *CloudTrie*. In Section V, we make experiments to compare our method with the state-of-the-art. Section VI reviews the related work. Section VII concludes this article with a summary.

II. Preliminary and Problem Statement

In this section, we first introduce the attack mechanism of prefix hijack. Next, we introduce three BGP-related datasets, which are used to construct *CloudTrie*.

A. Prefix Hijack Model

Based on the types of hijacked prefixes, these attacks can be further classified into prefix hijacking and sub-prefix hijacking [12]. Prefix hijacking occurs when an attacker forges an announcement for a prefix that is legitimately originated by another AS. The attacker then propagates this false information through BGP, causing other ASes on the Internet to mistakenly believe that the attacker's path is the best route to that prefix. As a result, traffic intended for the legitimate prefix can be intercepted or manipulated by the attacker, as illustrated in Fig. 1(a).

Sub-prefix hijacking occurs when an attacker announces a more specific prefix than the legitimate one, thereby exploiting the longest prefix match rule in BGP routing to alter the direction of traffic. For sub-prefix hijack, as shown in Fig. 1(b), a legitimate origin, AS10, announces the IP prefix 192.10.128.0/22, while an attacker, AS40, announces a more specific prefix, 192.10.128.0/24. According to the longest prefix match principle in the BGP route selection policy, BGP routers will select the path to the more specific sub-prefix. Consequently, traffic from AS20 and AS30 originally destined for AS10 is hijacked by AS40.

B. BGP-Related Routing Data

We introduce the BGP-related data, including three data sources, i.e., BGP routing information base (RIB), resource public key infrastructure (RPKI), and Internet routing registry (IRR). These three data sources are described in detail as below.

BGP RIB is a routing information base of BGP collectors, containing BGP routing information learned from neighbors [13]. This RIB data contains the global routing information; therefore, the data volume is vast. Although numerous BGP data collectors have been deployed worldwide, the collected data is incomplete and contains some misconfigured and malicious data [9].

Among the many collection points, this article selects the RRC00 for observation because it obtains a relatively complete routing table from ISPs and has a large number of peering neighbors [14].

RPKI is a BGP security extension developed by the Internet Engineering Task Force (IETF). This technique is based on public key infrastructure (PKI) technology and includes a cryptographic infrastructure for authenticating IP prefixes and ASes. A route origin authorization (ROA) record is a digital object issued by a prefix holder to authorize and validate the routing origin of an IP address prefix. It declares that a specific AS is the legitimate origin for its prefix, which helps prevent attacks such as prefix hijacking [15].

The IRR is a system composed of a set of distributed databases publicly used to store routing policies, prefixes, and AS association information submitted by network operators. The system is initially established to assist with the configuration of BGP routing policies [16].

C. Problem Statement

In the following, we give an intuitive example for stating the problem of the existing methods and why our method works. Upon receiving an update message, operators typically consult an information repository that functions similar to an allowlist to determine whether the message is malicious. The method first extracts the P/O pair from the message and checks it against the entries in the repository. If a match is found for both the prefix and the corresponding AS number (ASN), the method recognizes the update as benign. However, if the ASN corresponding to the prefix does not match the one in the repository, an alarm will be triggered. Furthermore, if the prefix is not found in the information repository, the method cannot determine whether the message is malicious.

Therefore, the accuracy of the information repository is crucial for detecting prefix hijacking. However, the repository

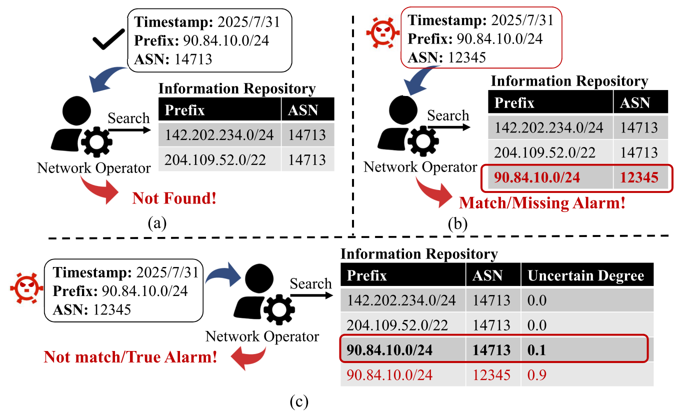

Fig. 2. Examples for comparison of the traditional method and our method. (a) Data missing. (b) Data inaccuracy. (c) Our method.

TABLE I
SUMMARY OF THE USED DATASET IN MEASUREMENT

Data Source	Collection Point	Data Volume
BGP RIB	RRC00	163GB
RPKI	RIPE, AFRINIC, APNIC, LACNIC, ARIN	7.0GB
IRR	RADB	77GB

collected from the BGP vantage points tends to be incomplete and inaccurate, which hurts the accuracy of AD.

In Fig. 2(a), if the information repository does not contain the P/O pair $\langle 90.84.10.0/24, 14\,713\rangle$ due to the incomplete data, the operator cannot match this item, and tends to be viewed as a potential anomaly. In Fig. 2(b), given a vicious update message, if the information repository contains an error record, i.e., $\langle 90.84.10.0/24, 14\,713\rangle$, operator can match this P/O pair with the information repository and thus mistakenly recognize it as correct, leading to a missing alarm.

Thus, we make the two following efforts to overcome data uncertainty. To construct a relatively comprehensive information repository, we attempt to collect most of the BGP-related data, encompassing not only the BGP RIB but also the sources of IRR and RPKI. Besides, to determine which P/O pairs are accurate, we compute the uncertain degree for each pair, shown in Fig. 2(c). The P/O pair $\langle 90.84.10.0/24, 14713\rangle$ has a lower uncertain degree than $\langle 90.84.10.0/24, 12345\rangle$, suggesting that $\langle 90.84.10.0/24, 14713\rangle$ is more credible, thus our method applies $\langle 90.84.10.0/24, 14713\rangle$ to make a match, finally obtaining the true alarm. In this way, the operator can not only build a more comprehensive repository, but also further evaluate confidence in each detection result according to the computed uncertain degree.

III. UNCERTAINTY MEASUREMENT OF BGP ROUTING DATA

In this section, we try to gain insight into the BGP uncertainty and measure the uncertainty of three BGP sources, i.e., BGP RIB, IRR, and RPKI. During the measurement process, we collected the data from the data sources during the period between January 2024 and December 2024, as shown in Table I.

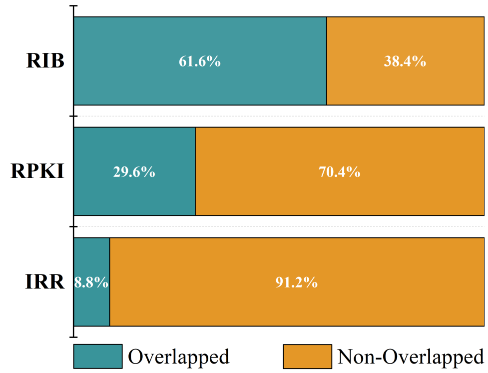

Fig. 3. Incomplete measurement on the BGP-related data.

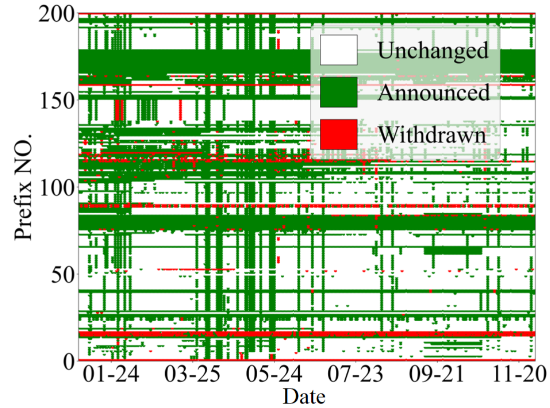

Fig. 4. Temporal variability of BGP prefixes.

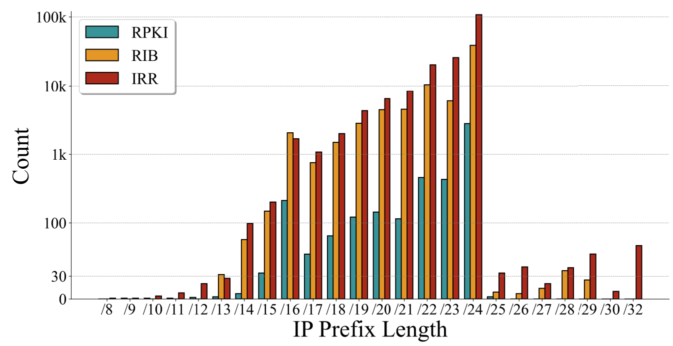

Fig. 5. Distribution of MOAS with the length of the prefixes.

A. Data Incompleteness

Given the same observation period, no single data source fully covers all global prefixes [17], named data incompleteness in this article. The incompleteness is mainly due to the locally deployed infrastructure for data collection. To quantify the extent of this incompleteness, we first collect the P/O pair set of each data source and then take the union of these sets to form the universal set. For each source, we calculate the proportion of its P/O pairs relative to this universal set. As shown in Fig. 3, RIB data covers the largest fraction of P/O pairs, followed by RPKI; IRR data provides the least coverage. Nevertheless, none of the three sources achieves complete coverage. This inherent data incompleteness limits the ability of existing detection methods to learn comprehensive routing patterns, thereby degrading their effectiveness.

B. Temporal Variability

For a given data source, routing information changes over time. The global inter-domain routing system is a distributed network where routing policy updates from different ASes and changes in network topology cause P/O pair relationships to exhibit significant temporal variability. In this article, we randomly selected 200 IP prefixes from BGP route update tables and tracked their announcement and withdrawal messages for one year, as shown in Fig. 4. The results suggest that different prefixes exhibit distinct patterns of change. Frequent BGP prefix announcements and withdrawals over time reflect significant volatility in inter-domain routing. This temporal variability of P/O pairs presents a major challenge for prefix hijacking detection. Therefore, the key lies in promptly adapting to dynamic routing changes and establishing a temporally stable knowledge base of P/O relationships.

C. Multiple Origin AS

In the inter-domain routing network, a single prefix is sometimes associated with multiple legitimate ASes, referred to as MOAS. MOAS is typically due to multiple BGP route announcements or path selections among different ASes. For instance, some large ISPs may share the same prefix, or an enterprise network may adopt various paths to enhance redundancy and reliability, resulting in the same prefix having multiple legitimate ASes. Thus, the detection method requires identifying which MOAS is benign or not. The related research [18] shows that the frequency and duration of MOAS events are significantly diverse and exhibit different distributions at various prefix lengths. As shown in Fig. 5, we analyze the number of MOAS conflicts with the prefix length varying in the BGP-related routing information. The analysis revealed that over 10% of prefixes exhibit MOAS behavior. The distribution of MOAS with different prefix lengths is similar for the three BGP sources. A large number of MOAS events are concentrated on prefixes of length 24, and the IRR data comprises more MOAS conflicts.

TABLE II
SUMMARY OF THE THREE DATA SOURCES CHARACTERISTICS

Data Source	# of P/O	Accuracy	MOAS	Time Variation
BGP RIB	High	Low	Medium	High
IRR	Low	Medium	High	Low
RPKI	Medium	High	Low	Medium

Based on the measurement results above, we summarize the characteristics of the three data sources in Table II. It can be observed that BGP data contains the richest information on P/O pairs, but its accuracy is limited. In contrast, RPKI data, while sparser in volume, offers significantly higher data quality.

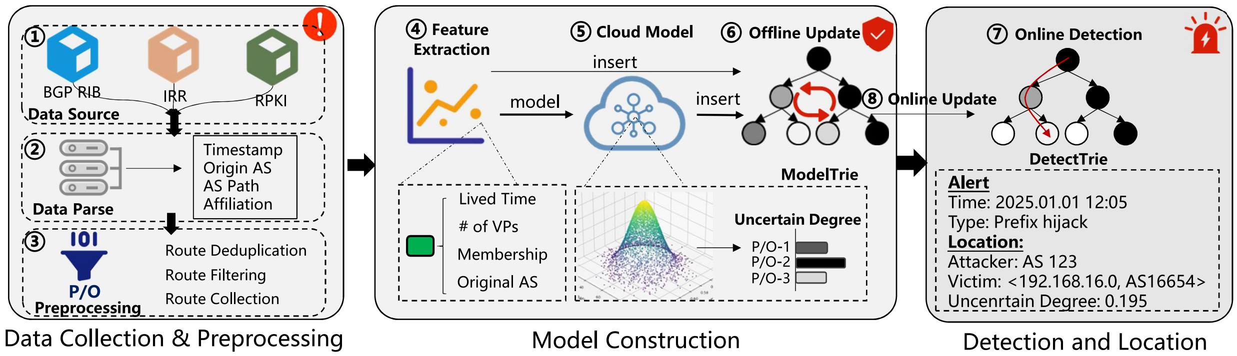

Fig. 6. Framework of *CloudTrie*, which comprises three modules, i.e., data collection and preprocessing, model construction, and detection and location.

TABLE III

RELATED FIELD INSERTED INTO MODEL TRIE

Field	Description
IP Prefix	The IP address prefix that the route needs to reach.
AS Path	The complete AS path from the source to the destination.
Timestamp	The time when the BGP message occurred.
Data Source	The source of data, i.e., BGP, RPKI, and IRR.
Original AS	The propagation origin of BGP updates.

IV. DESIGN OF CLOUDTRIE

To address the uncertainty in BGP routing data, this article designs a BGP AD method for uncertain data, with the framework illustrated in Fig. 6. We first conduct data extraction and preprocessing. Next, we extract the features of P/O pairs from three aspects, i.e., time, space, and data source. Exploiting this knowledge, we then construct a cloud model [19] for each pair to yield an uncertainty degree. Based on the uncertainty degree, the data is then filtered to build a set of trusted P/O mappings. Finally, the process concludes with AD and result evaluation. To overcome the temporal variability of BGP routing data, these mappings are updated automatically, ultimately enabling online detection.

A. Data Collection and Preprocessing

The collected raw data are generally compressed binary files. Therefore, we first need to preprocess the data, comprising data decompression parse to ensure the data is structured.

To reduce the distracting information existing in the parsed data, we clean up the BGP routes before model construction. First, the private ASNs appearing in the AS path are removed. Second, the aggregated AS paths are decomposed into multiple corresponding paths. Furthermore, we also remove the repetitive ASes on AS paths resulting from AS path prepending. Subsequently, we extract the key information from the specific field, including the IP prefix, path information, timestamp, and data source, where their particular meanings are listed in Table III.

B. Feature Extraction for P/O Pairs

It is crucial to extract spatio-temporally stable features that are robust to scene variations for the following model construction and AD. The related work has demonstrated that anomalous BGP data often exhibits significant temporal transience and spatial locality [3]. Therefore, to the best of our knowledge, we extract features of P/O pairs from the perspectives of time, space, and data source in advance to construct a spatial–temporal stability assessment model for these pairs. We introduce the definitions of the three proposed feature categories as follows.

Definition 1: To calculate the features of P/O pairs, we define the observation matrix $O \in \mathbb{N}^{c\times d}$ as follows:

$$O = \begin{bmatrix} o_{11} & \cdots & o_{1d} \\ \vdots & \ddots & \vdots \\ o_{c1} & \cdots & o_{cd} \end{bmatrix}_{c\times d} \tag{1}$$

where $o_{ij} \in \{0, 1\}$ indicates that if observation point i observes the target P/O pair during the jth time step, the value is 1; otherwise, it is 0. We denote c as the number of observation points and d as the length of the observation window. In the following, we try to characterize the certainty of P/O pairs from the perspectives of temporal persistence, spatial consistency, and data sources.

1) Temporal Persistence: Temporal persistence represents the duration for which an AS continuously announces a P/O pair. The longer the temporal persistence, the more stable the P/O pair is considered. Specifically, for a given P/O pair, the calculation is performed using the temporal persistence vector $h \in \{0, 1\}^{1\times d}$, which is defined as $h = [h_1, h_2, \ldots, h_d]$. Here, the element h_j is calculated as: $h_j = \max_{1\le i\le c} o_{ij}$. This implies that h_j equals 1 if at least one observation point observes the pair in the jth time step, and 0 otherwise. We use a weight vector $w_t \in \mathbb{R}^{1\times d}$ to represent the temporal weight coefficients within the observation window. The detailed calculation method is as follows:

$$s^t = \frac{h \cdot w_t^T}{\sum_{j=1}^{d} h_j}. \tag{2}$$

Dividing by the denominator ensures that P/O pairs exhibiting emerging stability are not assigned excessively low temporal persistence scores due to the observation window size.

2) Spatial Consistency: Spatial consistency represents the observed number of a P/O pair by the global vantage points. The higher the frequency of observation, the more spatially stable the P/O pair is considered. For a given P/O pair, the calculation is performed using the spatial consistency vector $g \in \mathbb{R}^{c\times 1}$. Here g is a column vector, where the ith element represents the total number of observations for the ith observation point across the entire time window. Specifically, $[g_1, g_2, \ldots, g_c]^T$, where each element g_i is the sum of the ith row of the observation matrix: $g_i = \sum_{j=1}^{d} o_{ij}$. We use a spatial weight vector $w_r \in \mathbb{R}^{c\times 1}$ to represent the observation point weights across different observation points

$$s^r = \sqrt{g^T \cdot w_r}. \tag{3}$$

3) Data Source: For each P/O pair, we also record its observed source. If one P/O pair can be followed by multiple data sources simultaneously, it indicates that the existence of the P/O pair is more stable and confident. We equip each P/O pair with a data source vector $k \in \mathbb{N}^{1\times 3}$. Here, $k_i \in \{0, 1\}$ indicates membership in a particular dataset (where $i = 1, 2,$ and 3 correspond to RPKI, IRR, and BGP RIB, respectively). If $k_i = 1$, it denotes the specific source views the P/O pair, while $k_i = 0$ denotes absence. The weight vector $w_m \in \mathbb{R}^{1\times 3}$ represents the degree of data reliability. Weights are assigned using a decay function based on the data quality of each source. The feature from the data source can be estimated as follows:

$$s^m = k \cdot w_m^T. \tag{4}$$

Based on the above three considerations, a quantitative uncertain degree of the spatio-temporal stability of P/O pairs can be estimated as follows.

C. P/O Cloud Model

The cloud model theory [19] is a mathematical tool that integrates probability theory and fuzzy mathematics to assess randomness and fuzziness. In this article, we model the randomness and fuzziness in the network using the cloud model to compute the uncertainty degree of each P/O pair.

Let a qualitative concept C be a concept on a quantitative universe of discourse U. If $x \in U$ is a random realization of the concept C, then the certainty degree of x with respect to C, $\mu(x) \in [0, 1]$, is a random number with a stable distribution: $\mu(x) : U \rightarrow [0, 1]\ \forall x \in U$. The distribution of x on the universe U is defined as a cloud model and x represents a cloud drop belonging to the universe U. Deriving the core parameters from cloud drops $x \in U$ constitutes the cloud construction process. Conversely, determining the certainty degree of a cloud drop via the cloud model is referred to as the cloud reasoning process. The entire cloud process integrates probability theory and fuzzy set theory, thereby effectively characterizing the fuzziness and randomness of the concept [20].

Definition 2: A P/O cloud is a qualitative concept for the stability of a P/O pair represented by a cloud model, and it is an extension of a 1-D normal cloud. The cloud model is represented by several known normal cloud drops $\bar{x}_i = (s_i^t, s_i^r, s_i^m)$, where the three elements of the vector $\bar{x}_i$, respectively, represent the temporal persistence, spatial consistency, and data source feature components of the cloud drop $\bar{x}_i$.

For the P/O pair feature cloud, each cloud model comprises three components, represented as $C((Ex_t, Ex_r, Ex_m), (En_t, En_r, En_m), (He_t, He_r, He_m))$. The construction process is detailed below.

1) Expectation: Expectation (Ex) is the most typical sample value in the quantification of a qualitative concept. For a given set with a cloud drops, $\{(s_1^t, s_1^r, s_1^m), \ldots, (s_a^t, s_a^r, s_a^m)\}$, the estimate value of the expectation Ex_t is calculated as follows:

$$\hat{E}x_t = \bar{S}_t \tag{5}$$

where $\bar{S}_t$ represents the mean of s^t, calculated as follows:

$$\bar{S}_t = \frac{1}{a}\sum_{i=1}^{a} s_i^t. \tag{6}$$

The calculation for the spatial consistency and data source feature components, $\bar{S}_r$ and $\bar{S}_m$, is similar to (6).

2) Entropy: Entropy (En) is determined by the randomness and fuzziness of a qualitative concept. Thus, it is used to measure the degree of dispersion of all cloud drops and also reflects the range of values that the cloud drops accepted by the concept can take. For the P/O pair cloud with a drops, we estimate the value of En_t as follows:

$$\hat{E}n_t = \sqrt{\frac{\pi}{2}} \times \frac{1}{a}\sum_{i=1}^{a}\left(s_i^t - \bar{S}_t\right). \tag{7}$$

3) Hyper-Entropy: Hyper-entropy (He) is used to measure the uncertainty of entropy and describes the correlation between fuzziness and randomness. Based on this concept, the estimated hyper-entropy value for each feature component can be calculated separately. The estimated hyper-entropy for the temporal persistence component is

$$\hat{H}e_t = \sqrt{B_t^2 - \hat{E}n_t^2} \tag{8}$$

where B_t is the sample variance of feature component t, and $B_t^2 = 1/(a-1)\sum_{i=1}^{a}(s_i^t - \bar{S}_t)^2$.

In the cloud model theory, a multidimensional cloud model is constructed based on P/O drops from three perspectives: temporal persistence, spatial consistency, and data source reliability; this procedure is referred to as the cloud construction process. Furthermore, the certainty degree of a P/O pair can be computed using the membership degree of the corresponding cloud drop within the established cloud model. Consequently, in this article, the uncertainty of a P/O pair is defined as follows.

Definition 3: Given the universe of discourse $U = \{\bar{x}_i | \bar{x}_i = (s_i^t, s_i^r, s_i^m), i \in \mathbb{N}\}$, the certainty degree for a cloud drop $\bar{x}_i$ with respect to the feature cloud C is calculated as follows:

$$\hat{y}_i = \left[e^{-\frac{\left(s_i^t - Ex_t\right)^2}{2(En_t')^2}} \quad e^{-\frac{\left(s_i^r - Ex_r\right)^2}{2(En_r')^2}} \quad e^{-\frac{\left(s_i^m - Ex_m\right)^2}{2(En_m')^2}} \right] \cdot \mathbf{W}^T. \tag{9}$$

In (9), $\mathbf{W} \in \mathbb{R}^{1\times 3}$ is the weight vector for the three dimensions. En_t' represents a normally distributed random variable generated from a normal distribution $N(En_t, He_t^2)$, where En_r' and En_m' are defined analogously. Each exponential term effectively serves as a fuzzy membership function, mapping the feature deviation to a certainty score $\hat{y}_i \in [0, 1]$. This metric quantifies the similarity between the observation and

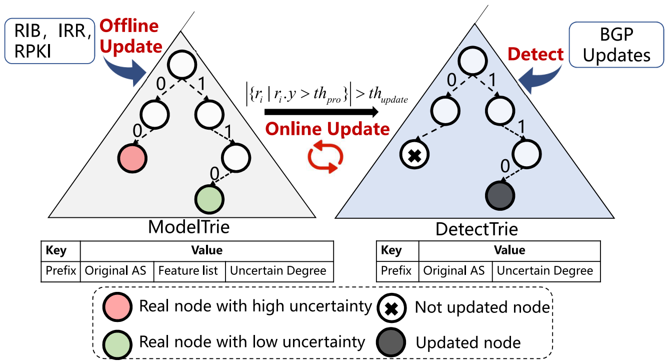

Fig. 7. Framework of AD and update mechanism.

the "normal state cloud," encapsulating the transformation from quantitative values to qualitative concepts.

Furthermore, the degree of uncertainty for P/O pair i, denoted as $\hat{u}_i$, is estimated as follows:

$$\hat{u}_i = 1 - \hat{y}_i. \tag{10}$$

If the calculated probability $\hat{u}_i$ for a P/O pair x_i exceeds a certain threshold, th_{pro}, it denotes that the pair is unstable. This P/O pair is incorrect when used in AD with a confidence of $\hat{u}_i$. Therefore, we remove it and exclude it from the AD.

D. AD and Model Update

We use the Trie data structure to store the related data and carry out AD. Two Tries are organized to detect anomalies and model update, i.e., ModelTrie and DetectTrie. ModelTrie is used to store IP prefixes as keys, and at the corresponding nodes, it also stores the origin AS, feature vector, and the calculated uncertain degree. DetectTrie stores P/O pairs that have a high degree of certainty for online detection. The two trees have different roles in the proposed method and work collaboratively to improve detection efficiency and accuracy, with the specific framework shown in Fig. 7.

ModelTrie: This trie stores well-extracted feature vectors to construct a cloud model and evaluates the uncertainty degree, i.e., $\hat{u}_i$, for all prefixes. When a new P/O pair appears, the system evaluates its $\hat{u}_i$ in real-time within the ModelTrie based on the cloud model.

DetectTrie: We use this trie for online AD. It stores P/O pairs with an uncertainty degree, i.e., $\hat{u}_i$ below a certain threshold. DetectTrie is used to determine whether a prefix corresponds to a legitimate route rapidly and to provide an interpretable report upon AD.

1) Anomaly Detection: In the detection phase, for each incoming route update, the system utilizes prefix covering and matching rules to evaluate the relationship between the incoming route and the legitimate prefixes stored in the DetectTrie. It verifies whether the incoming prefix is legitimate and determines if it matches a legitimate ASN. If a mismatch is found, *CloudTrie* marks this update message as an anomaly and triggers further alerts, as well as generates an anomaly report. Additionally, the uncertainty degree of the P/O pair, $\hat{u}$, is used to represent the level of uncertainty in the AD.

In the following description, the prefix q and origin ASN a_o represent a P/O pair stored in the DetectTrie, while the route prefix and origin ASN from the update message are r and a_u, respectively. In accordance with the RFC 6811 [21], the following two operations are defined.

Cover(q, r): If the length of prefix q is less than or equal to the length of route prefix r, and the prefix q and the address of route prefix r are identical for all bits specified by the length of prefix q, then the prefix r is said to be covered by the prefix q.

Match(q, r): A match occurs when route prefix r is covered by prefix q, the length of route prefix r is less than or equal to the maximum length specified for prefix q, and the route origin ASN is equal to the origin AS in the P/O pair. This can be expressed by the following equation:

$$\text{Match}\,(q, r) = \text{Cover}\,(q, r) \wedge \text{Cover}\,(r, q) \wedge (a_o == a_u). \tag{11}$$

The online AD is carried out by the following process.

Valid: A prefix q is considered valid if and only if it conforms to the known legitimate routes (via Cover$(\cdot)$ and Match$(\cdot)$ conditions). This can be expressed by the formula

$$\text{Valid}\,(q) = \text{Match}\,(q, r). \tag{12}$$

Invalid: If a prefix does not conform to any legitimate routing rule (i.e., it is covered or mismatched), it will be considered invalid and illegitimate. When route prefix r covers prefix q, and prefix q covers route prefix r, but the ASN is not equal to a_o, and the match fails, it is considered a potential prefix hijacking incident. This process can be expressed by the formula

$$\text{Hijack}\,(q) = \text{Cover}\,(q, r) \wedge \text{Cover}\,(r, q) \wedge \neg\text{Match}\,(q, r). \tag{13}$$

When prefix q covers route prefix r, meaning route prefix r is more specific than prefix q, and the announced ASN is not equal to the origin AS, it is considered a potential sub-prefix hijacking incident. Sub-prefix hijacking is defined as a malicious attacker carrying out route hijacking by constructing the most specific BGP route message. It can be formalized as

$$\text{Sub\_Hijack}\,(q) = \text{Cover}\,(q, r) \wedge \neg\text{Match}\,(q, r). \tag{14}$$

The membership degree $\hat{y}_q$ of the P/O pair is used as the certainty level for AD.

Not Found: When the announced prefix is not represented in the current decision tree, no matching route is found. This can be expressed by the formula

$$\text{NotFound}\,(q) = \neg\text{Cover}\,(q, r). \tag{15}$$

This situation indicates that the prefix is newly emerged and has not appeared in any data sources (RIB table, RPKI, and IRR). Therefore, the prefix is marked as pending verification.

2) Offline Update: It is used to update ModelTrie by continuously extracting features related to temporal persistence, spatial consistency, and data sources from streaming BGP routing information. Update the cloud model and ModelTrie according to the update frequency of routing data (IRR, RPKI, and BGP RIB).

3) Online Update: It is used to update DetectTrie. In dynamic network environments, the mapping relationships of P/O pairs change dynamically. To ensure the accuracy of P/O pairs within the DetectTrie and prevent overly frequent updates, we apply lazy update mechanism, and set a threshold $\text{th}_{\text{update}}$ designed as follows: when the number of newly added P/O pairs exceeds a certain threshold $\text{th}_{\text{update}}$ (i.e., $|\{r_i \mid r_i.y > \text{th}_{\text{pro}}\}| > \text{th}_{\text{update}}$), the proposed method automatically updates the DetectTrie based on the ModelTrie.

Through these update mechanisms, the ModelTrie and DetectTrie can cooperate continuously to verify the prefix and update the model, thereby enhancing the real-time performance and accuracy of detection.

E. Complexity Analysis

The complexity of *CloudTrie* primarily consists of two parts: model construction and AD, which are analyzed separately below.

Model construction: Primarily involves data preprocessing, feature extraction, and calculating key cloud model parameters. For data preprocessing, the raw routing data (RIB, RPKI, and IRR) undergoes an initial preparation phase, which has a time complexity of $O(M)$, where M represents the total number of entries in the original dataset. Feature extraction breaks down as follows: temporal persistence calculation takes $O(1)$ for a single computation, leading to an overall $O(M)$ complexity when iterating through all M P/O pairs; spatial consistency calculation requires traversing C observation points for each computation, resulting in an $O(C)$ complexity per calculation and an overall $O(M \cdot C)$ for all M P/O pairs; data source calculation is constant time, $O(1)$, per computation, summing up to $O(M)$ for all P/O pairs. Finally, for cloud model calculation (forward cloud transformation), the normal cloud parameters (Ex, En, He) are computed for each of the three feature components (temporal, spatial, and membership degree) of every P/O pair. Each feature's parameter calculation has a complexity of $O(M)$, leading to an overall complexity of $O(3M)$, which simplifies to $O(M)$.

Anomaly detection: This process primarily involves the operation of node insertion and searching in Trie. Insertion and search operations for each IP prefix have a time complexity of $O(L)$, where L is the prefix length ($L \leq 32$ for IPv4). Searching through either the ModelTrie or DetectTrie has a time complexity of $O(L)$.

Update mechanism: When the dynamic update threshold is triggered, only local adjustments to the Trie structure are required, resulting in a time complexity of $O(P \times L)$, where P denotes the number of prefixes.

V. Experiment and Evaluation

In this section, we carry out extensive experiments to validate our method compared with the state-of-the-art methods. The experimental results suggest that *CloudTrie* can detect an average of 117,647 route update messages per second while maintaining low memory usage. In addition, CloudTrie can detect all of events in the dataset, outperforming the other methods.

TABLE IV
Descriptions of Closed Dataset

NO.	Event	Start Time	End Time
1.	hijack-20151106-BHARTI	2015/11/06 05:52	2015/11/06 14:40
2.	hijack-20151204-Volu	2015/12/04 19:30	2015/12/06 21:00
3.	hijack-20170105-Iran	2017/01/05 12:00	2017/01/06 18:00
4.	hijack-20170426-PJSC	2017/04/26 22:36	2017/04/26 22:43
5.	hijack-20171212-DV	2017/12/12 04:43	2017/12/12 04:46
6.	hijack-20180629-Bitcanal	2018/06/29 13:00	2018/06/29 22:00
7.	hijack-20180730-Iran	2018/07/30 10:30	2018/08/05 12:00
8.	hijack-20190508-FIBRA	2019/05/08 15:08	2019/05/09 15:11
9.	hijack-20210205-Campana	2021/02/05 15:51	2021/02/05 16:51
10.	hijack-20220215-Nigeria	2022/02/15 13:30	2022/02/15 14:30
11.	hijack-20220228-RU	2022/02/28 17:00	2022/02/28 23:59
12.	hijack-20220328-RU	2022/03/28 12:05	2022/03/28 12:50

A. Datasets Description

We used multisource datasets, including RPKI, IRR, and RIB tables, to carry out the experiment. For the RIB and BGP Updates, we selected the representative collection point RRC00 from RIPE RIS. We employed the BGPdump tool to parse the MRT-formatted files and extract key fields, including Prefix, AS-PATH, and Peer. For the IRR data, we extracted autnum and route objects from the public databases of the five Regional Internet Registries (RIPE, APNIC, ARIN, LACNIC, and AFRINIC) to construct P/O pairs. For the RPKI data, we utilized historical archives from RPKIviews to obtain validated ROA objects, serving as the trusted routing baseline. Subsequently, the prefixes from these three data sources were unified into a binary format and inserted into trie structure using a sliding window mechanism. Within this structure, each AS node stores a P/O observation matrix along with data source indicators. The experimental data in Sections V-E, V-F, and V-G were tested using data from the second half of 2024 (from July to December), totally 3.1 TB. The closed experimental dataset in Section V-J contains 12 prefix hijacking events, with specific details shown in Table IV. Each BGP update message was labeled by matching the victim AS, prefix, and attacker AS and the matched messages were annotated as anomalous. This labeled dataset amounts to 220 GB, comprising approximately 1.5 billion BGP update messages. The open dataset in Section V-K was tested using data from the second half of December 2024.

B. Implementation

The method and the baselines are implemented in Python and run on a Linux operating system with an AMD EPYC 7K62, 48-core processor and 64GB of RAM. To ensure fairness, both the proposed method and the compared methods use the identical setup.

In real-world network management scenarios, the number of alerts and false alarm are the most concerned metrics for operations and maintenance personnel. Thus, we choose them to evaluate the detection accuracy of each method.

We use the detection time, denoting the period from receiving an update message to producing a detection result, as the evaluation metric in terms of time efficiency.

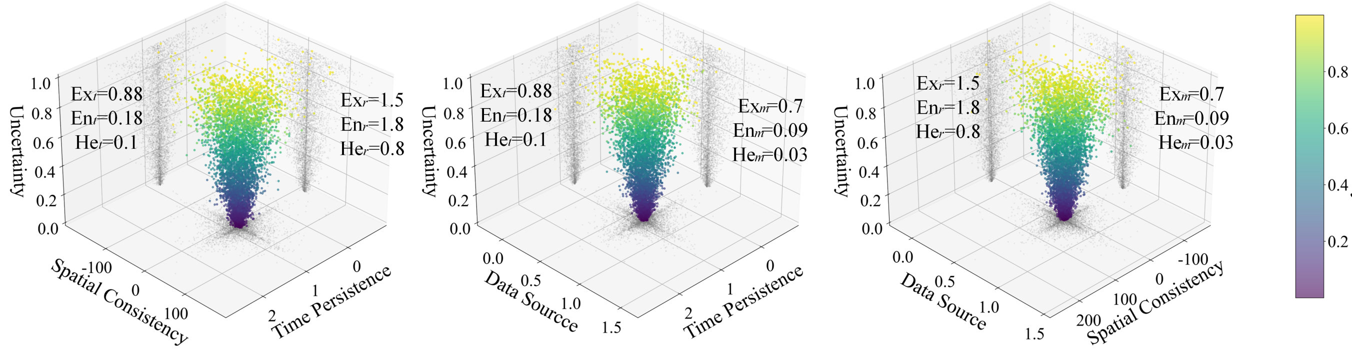

Fig. 8. Cloud model projection under any feature pairs.

TABLE V

DESCRIPTION OF BASELINES

Method	Category	Key Technologies
Artemis [22]	Information database	Legitimate route comparison
BEAM [2]	Machine Learning	Node Embedding
BGPviewer [23]	Machine Learning	Graph attention network
BGPvector [24]	Machine Learning	Skip-gram

TABLE VI

DETAILS OF HYPERPARAMETERS BY THE GRID SEARCH

Method	Grid Search Hyperparameters
BGPvector	Emb [32, 64, 128], Neg [5, 10], Dense [100, 200]
BEAM	Emb [64, 128, 256], Neg [5, 10], Lr [1e-4, 1e-3, 1e-2]
BGPviewer	Window [8, 16, 32], Neg [5, 10], Hid [64, 128, 256]

C. Baselines

In the experiments, we compare our method with the state-of-the-art methods, comprising Artemis, BEAM, BGPviewer, and BGPvector, which can be used to detect prefix hijack. The details of the baseline methods are listed in Table V. In addition, the key hyperparameters of each method have been optimized using grid search to maximize performance. The details of these hyperparameters are listed in Table VI.

D. Key Questions

In the experiments, we will answer the following key questions (KQs) about our method *CloudTrie*. KQ1: What does this constructed P/O cloud model look like? KQ2: How do the key parameters of *CloudTrie* change adaptively over time? KQ3: Facing vicious injection, how robust is *CloudTrie*? KQ4: How does *Cloudtrie* select the threshold in real deployment? KQ5: How does *CloudTrie* behave in real-world anomaly events? KQ6: How does *CloudTrie* behave in an open dataset with a continuous data stream?

E. P/O Cloud Model (KQ1)

As shown in Fig. 8, we present the projection of cloud model onto any two feature dimensions. Each cloud drop represents the spatio-temporal features of a P/O pair, with its color intensity indicating the degree of uncertainty of that P/O drop (9). Fig. 8 shows a temporal expectation (Ex_t) of 0.88 and a spatial expectation (Ex_r) of 1.5, suggesting that most of the P/O pairs exhibit high spatio-temporal stability. The temporal entropy (En_t) is 0.18, while the spatial entropy (En_r) is 1.8. This indicates lower data volatility in the temporal dimension, whereas the spatial dimension shows higher dispersion due to variations in observation point distribution. Furthermore, $He_t = 0.1$ and $He_r = 0.8$ denote good certainty for both temporal and spatial parameters. The feature parameters across these three dimensions vary, allowing the constructed cloud model to analyze the stability of P/O pairs from different perspectives.

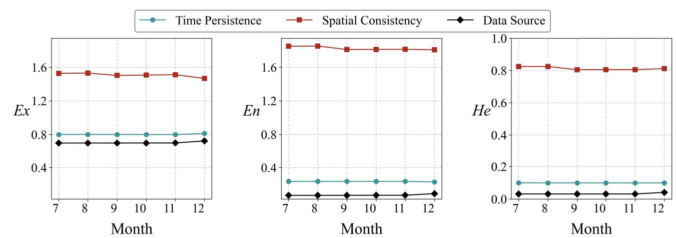

Fig. 9. Dynamic trend of cloud model parameters (2024).

F. Time Variation of CloudTrie (KQ2)

To validate the cloud model's adaptability to dynamic BGP routing changes, this experiment analyzes the temporal evolution of the cloud model's core parameters (Expectation Ex, Entropy En, and Hyper-entropy He). This analysis aims to evaluate the model's dynamic adaptability. The experiment employs a sliding time window mechanism (with a window size of 5 days), and the results are presented in Fig. 9.

1) *Expectation (Ex):* The Ex value in the temporal persistence dimension remains stable throughout the observation period, with fluctuations significantly smaller than those in traditional models. This indicates a high degree of consistency in the temporal stability characteristics of P/O pairs. Although the Ex value in the spatial consistency dimension shows a periodic fluctuations due to variations in observation point distribution, its

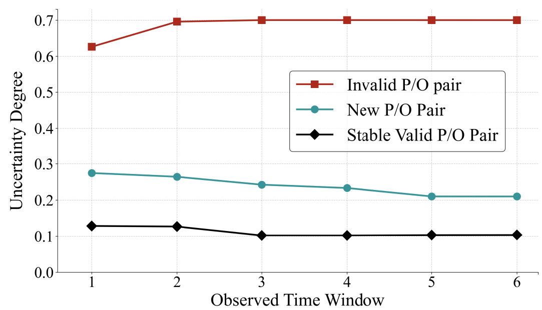

Fig. 10. Stability observation of P/O pairs.

standard deviation is still strictly maintained at a low level. The results verify the multi-source data fusion mechanism's ability to suppress spatial noise.

2) *Entropy (En):* The *En* value across all 3-D remained stable throughout the observation period, with the most prominent decrease observed in the data source dimension. This trend suggests that, over time, the model progressively reduced inconsistencies among data sources through dynamic adjustments, thereby enhancing the robustness of deterministic judgments.
3) *Hyper-Entropy (He):* The *He* values for all dimensions were consistently constrained to an extremely low range. Over time, significant changes were observed in both spatial consistency and data source dimensions.

Experimental results demonstrate that the proposed method effectively captures the intrinsic patterns of time-varying routing data by continuously optimizing the statistical characteristics of the feature clouds, thereby ensuring the dynamic adaptability of *CloudTrie*.

G. Robustness Analysis (KQ3)

To measure the robustness of the proposed method, we observe three categories of P/O pairs (1000 samples for each category): stable valid P/O pairs, newly added valid P/O pairs, and invalid P/O pairs. Fig. 10 illustrates the variation of the uncertainty degree $\hat{u}_i$ for these three categories during December 2024. It can be observed that the uncertainty degree of invalid P/O pairs consistently remains significantly higher than that of both newly added valid P/O pairs and stable P/O pairs. Meanwhile, the uncertainty of newly added valid P/O pairs was slightly higher than that of stable P/O pairs initially, before decreasing slightly and largely remaining stable below 0.2. It indicates that invalid P/O pairs consistently maintain a high degree of uncertainty, while stable pairing relationships maintain a low degree of uncertainty within their respective periods, demonstrating good stability. Notably, this distinct separation in uncertainty characteristics verifies the model's adversarial robustness against poisoning attacks. An intelligent adversary might attempt to manipulate the Cloud Model parameters by injecting low-frequency, seemingly harmless BGP updates that fall between the characteristics of Invalid and New P/O pairs. However, since the Reverse Cloud Generator algorithm predominantly relies on high-quality cloud drops generated from Stable Valid P/O pairs, these adversarial

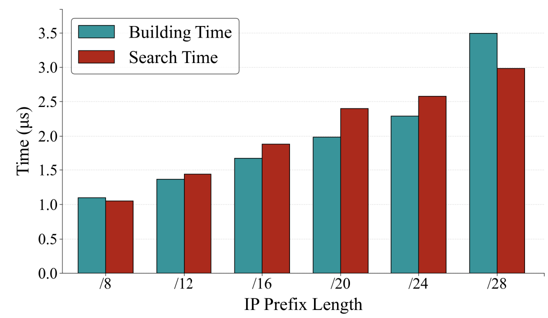

Fig. 11. *CloudTrie* construction and detection speed with various prefix lengths.

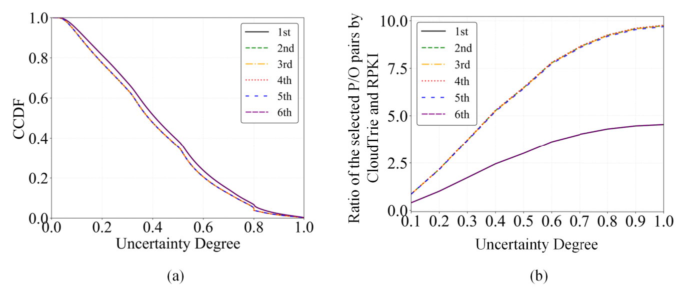

Fig. 12. (a) P/O pairs confidence CCDF curve in January 2024. (b) Stabilize the dynamic correlation between the number of P/O pairs and the uncertainty threshold.

data points are statistically filtered out. Consequently, the malicious updates fail to accumulate enough weight to shift the Expectation, making it extremely difficult for attackers to alter the model's decision boundary. Fig. 11 shows the construction and search times for the DetectTrie as prefix length varies. This demonstrates that the proposed method is capable of performing real-time AD across different prefix lengths.

The experimental results demonstrate that the cloud model proposed in this article can effectively distinguish between legitimate and illegitimate P/O pairs. Furthermore, it remains robust to the negative influence of illegitimate P/O pairs.

H. Distribution of Uncertainty Degree (KQ4)

In this section, we collect all of the P/O pairs within an observation window (the window size is set to five days) and further calculate the distribution of uncertainty degree of P/O pairs. Then, we collect the results and depict them in Fig. 12. Additionally, we evaluated the cloud model within a real-world network environment using a dataset of 300 k P/O pairs, which incorporates nearly 3000 anomalous pairs from the same temporal window. The evaluation outcomes are illustrated in Fig. 13.

Fig. 12(a) displays the complementary cumulative distribution function (CCDF) of the uncertainty degree. When the uncertainty degree is 0.4, the CCDF values of all curves drop significantly. This indicates that if an uncertainty threshold of 0.4 is selected, 50% of the P/O pairs can be retained, and the uncertainty degree of these P/O pairs is within 40%.

Fig. 12(b) uses RPKI's P/O pairs as a baseline to show the ratio of P/O pairs selected by *CloudTrie* to those in RPKI. As

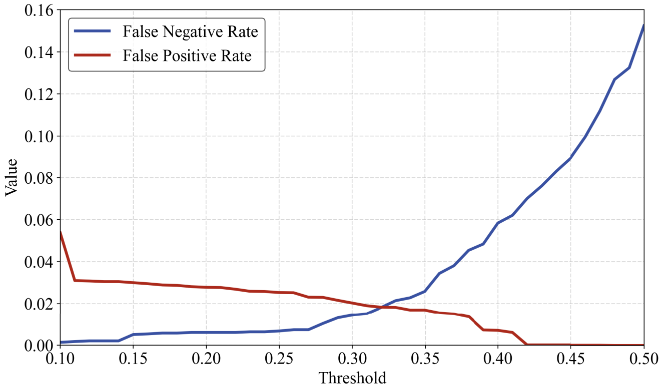

Fig. 13. Performance of cloud model across varying threshold settings.

the threshold increases from 0 to 1, the number of stable P/O pairs rises, then flattens out as the threshold approaches 1. It can be observed that the P/O pairs chosen by *CloudTrie* are two to ten times more numerous than RPKI's P/O pairs. It indicates that the selected P/O pairs can effectively supplement the RPKI data source and provide an uncertainty degree for each P/O pair.

I. Sensitivity Analysis

The performance was assessed using false positive rate (FPR) and false negative rate (FNR), as illustrated in Fig. 13. The results demonstrate an inverse relationship: as the threshold is raised, the FPR diminishes, whereas the FNR increases. This trend suggests that a higher threshold allows for the inclusion of more legitimate P/O pairs that exhibit some uncertainty. Notably, at an uncertainty threshold of approximately 0.4, almost all normal P/O pairs are added to Detecttrie. However, this relaxation also permits a small number of anomalous P/O pairs to enter the system as the threshold increases further. Finally, we selected a threshold th_{pro} of 0.4 for the experiment. This setting ensures that 50% of the data is retained with a high degree of certainty, while simultaneously maintaining low FPR and FNR. Notably, the volume of this retained data is nearly five times larger than that of the RPKI data. In this field, minimizing the FNR is more crucial than reducing the FPR. This means that network operators can take more time in further anomaly mitigation rather than filtering out anomalies after detecting anomalies.

In the real deployment, we apply a similar mechanism to update the threshold when carrying out each time of online detection.

J. Performance Comparison in Closed Datasets (KQ5)

In this section, we evaluate the accuracy of our proposed method using a closed environment, comprising 12 labeled real-world events. The experimental results are listed in Table VII.

Compared to the other methods, *CloudTrie* demonstrates its superior performance by detecting all of the anomaly events with the lowest number of false positives. This efficacy stems from its ability to integrate data from three distinct

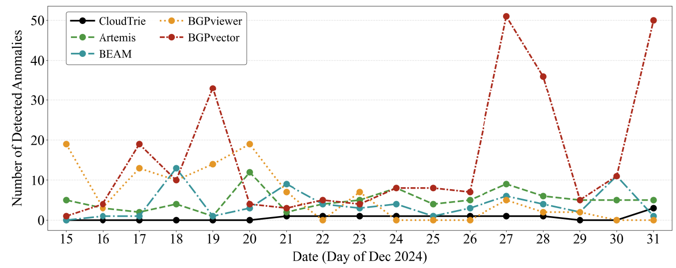

Fig. 14. Comparison results of different AD methods under the open environment.

sources and filter out stable P/O pairs, which in turn leads to more accurate AD results. Instead, the other methods neither detect all of events. This is because these methods do not have sufficient accurate data to model their knowledge system. Specifically, the method, like Artemis, which relies on information repositories, may generate false alarms when routing information fails to match, particularly if the repository data is not comprehensive. Similarly, the machine learning-based approaches may experience a reduction in detection accuracy and an increase in false alarms if their training data is insufficient.

K. Performance Comparison in Open Dataset (KQ6)

To further validate the robustness and adaptability of *CloudTrie* in a real-world scenario, we performed AD with the continuous BGP routing data during the period of the latter half of December 2024. The comparative performance of our proposed method against the baselines is presented in Fig. 14.

Fig. 14 reveals that *CloudTrie* exhibits a consistently low number of alerts, with an average of 0.65 alerts per day and a variance of 0.65, indicating stable performance. In contrast, BGPvector and BGPviewer had the highest and second-highest alert frequencies, respectively. BGPvector averaged 71.82 alerts per day with a variance of 399.97, while BGPviewer averaged 5.88 alerts per day with a variance of 41.95.

We also record the memory and detection time for each update message during their online detections, as shown in Table VIII. We can observe that *CloudTrie* behaves best in terms of detection time and takes up the second-lowest memory usage. *CloudTrie* strategically utilizes only three feature dimensions to construct the Cloud Model—despite the potential for superior performance with higher dimensions—to achieve an optimal balance between effectiveness and computational overhead. This stands in stark contrast to deep learning baselines, which incur massive costs by requiring the extraction of 26-D features for every sliding window. Furthermore, *CloudTrie*'s unsupervised paradigm fundamentally overcomes the critical bottlenecks of supervised learning: specifically, the scarcity of labeled BGP data, significant temporal discrepancies, and severe class imbalance. As highlighted by BEAM, these data characteristics frequently drive deep models toward gradient collapse and training failure, necessitating extensive parameter optimization; in contrast, *CloudTrie* completely circumvents such risks. Finally, unlike the opaque

TABLE VII
EXPERIMENTAL RESULTS UNDER THE CLOSED DATASET

Dataset		Detected					# Alarms (# False Alarm)				
No.	**Event**	**CloudTrie**	**Artemis**	**BEAM**	**BGPvector**	**BGPviewer**	**CloudTrie**	**Artemis**	**BEAM**	**BGPvector**	**BGPviewer**
1.	hijack-20151106-BHARTI	✓	✓	✓	✓	✓	**20 (1)**	10 (4)	25 (10)	6 (5)	17 (4)
2.	hijack-20151204-Volu	✓	✓	✓	✓	✓	**44 (3)**	74 (33)	30 (25)	16 (6)	19 (16)
3.	hijack-20170105-Iran	✓	✓	×	×	✓	**16 (0)**	**16 (0)**	17 (17)	25 (25)	4 (2)
4.	hijack-20170426-PJSC	✓	×	✓	✓	✓	8 (6)	0 (0)	8 (6)	**1 (0)**	**1 (0)**
5.	hijack-20171212-DV	✓	✓	✓	✓	×	**8 (0)**	**8 (0)**	8 (7)	3 (2)	0 (0)
6.	hijack-20180629-Bitcanal	✓	✓	×	×	✓	**7 (1)**	**7 (1)**	15 (15)	5 (5)	9 (0)
7.	hijack-20180730-Iran	✓	✓	✓	✓	×	**3 (0)**	**3 (0)**	12 (11)	1 (0)	0 (0)
8.	hijack-20190508-FIBRA	✓	×	×	×	✓	**4 (0)**	0 (0)	8 (8)	12 (12)	7 (6)
9.	hijack-20210205-Campana	✓	×	×	×	✓	3 (2)	4 (4)	8 (8)	2 (2)	**1 (0)**
10.	hijack-20220215-Nigeria	✓	✓	×	×	✓	**2 (1)**	**2 (1)**	10 (10)	2 (2)	3 (2)
11.	hijack-20220228-RU	✓	✓	×	×	✓	10 (9)	334 (323)	0 (0)	3 (3)	**2 (1)**
12.	hijack-20220328-RU	✓	✓	×	×	✓	**4 (0)**	6 (2)	9 (9)	2 (2)	10 (9)
	Overall	12/12	9/12	5/12	5/12	10/12	**129 (23)**	464 (368)	150 (126)	78 (64)	73 (40)

TABLE VIII
MEMORY AND AVERAGE DETECTION TIME IN OPEN TEST

Method	Memory Usage (MB)	Average Time
CloudTrie	34.3	8.5 μs
Artemis	26.2	9 μs
BEAM	150	15 ms
BGPviewer	667.5	50 ms
BGPvector	120	7 ms

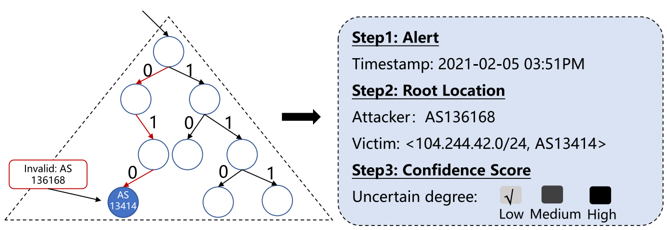

Fig. 15. Analysis of Myanmar prefix hijacking incident.

"black-box" nature of deep learning models, *CloudTrie* demonstrates superior interpretability.

L. Case Study

On February 5, 2021, Myanmar's Ministry of Transport and Communications issued a notice to the country's mobile network and ISPs, attempting to block Twitter and Instagram traffic through prefix hijacking [25]. Specifically, AS136168 (Campana MYTHIC) began announcing 104.244.42.0/24 at 15:51 UTC on February 5, despite 104.244.42.0/24 legitimately belonging to AS13414 (Twitter). Our proposed method, *CloudTrie*, successfully detect this event at the corresponding time. The whole process is presented in Fig. 15.

In this event, we identify that two distinct ASes, namely AS136168 and AS13414, were both announcing reachability to the same IP prefix, 104.244.42.0/24. Concurrently, *CloudTrie* assigns a low uncertainty degree to this incident ($\hat{u}_i = 0.195$), indicating a high level of confidence in the detection of this event.

VI. RELATED WORK

In recent years, researchers have extensively studied prefix hijacking detection, broadly categorizing approaches into the following three types.

A. Active Probing-Based Detection Methods [26], [27], [28]

This type of instrumentation works by injecting traffic into the network and then analyzing its reception and stability to determine the reachability of a specific IP address or prefix. For instance, BGP Beacon [26] periodically announces and withdraws special IP addresses. Beacon receivers monitor these updates to gauge inter-domain network stability and convergence. iSPY [27] actively sends probe traffic to target prefixes to detect their reachability and path changes. If probe results do not match the expected path, it might indicate a prefix hijacking or other routing anomaly. To further reduce FRRs, Schlamp et al. [28] utilize encrypted traffic based on SSL/TLS to test the reachability of anomalous IP addresses. By comparing public key changes before and after an anomalous event, they determine if it's a benign occurrence. In order to reduce the network overhead, Trinocular [29] applies an adaptive probing mechanism. Although their good ability of detecting anomalies, this type of methods aggravate the burden of network.

B. Machine Learning-Based Methods [30], [31], [32]

They use algorithms to model network traffic or routing data to identify anomalous patterns. For instance, Vanerio and Casas [32] propose an ensemble learning-based AD framework that combines supervised and unsupervised learning. This framework leverages multiple trained base classifiers and integrates them using unsupervised learning, improving detection accuracy and robustness [33]. The advantage of these methods lies in their ability to learn complex network behavior patterns automatically. However, machine learning approaches typically require large amounts of labeled data for training, and model performance heavily relies on data quality and the effectiveness of feature engineering [34]. Additionally, the "black-box" nature of deep learning models makes it difficult to interpret detection results, thereby limiting their trustworthiness in practical applications [35].

C. Information Repository-Based Methods [36]

These methods analyze BGP routing information according to an information repository comprising a number of valid pieces of information. Artemis is widely used for detecting prefix hijacking. The advantage lies in its strong interpretability, offering clear logical explanations for each detection decision. However, routing logic methods are highly dependent on the reliability of the data. If the input data sources are inaccurate or incomplete, detection results can be significantly affected. Especially when dealing with partially missing or inconsistent data, routing logic methods are prone to generating false alarms.

Unlike other methods, this article proposes a novel BGP AD approach fusing multiple uncertain information sources. Our primary aim is to build a detection instrumentation system that can collect more valid data from uncertain BGP data source to achieve more accurate AD. Compared to the existing methods, *CloudTrie* can detect hijacking behaviors more accurately, while also providing interpretable results.

VII. Conclusion

BGP route security is crucial for global Internet connectivity [37]. This article introduces *CloudTrie*, an efficient AD method that addresses the issue of uncertain information sources in BGP prefix hijacking by integrating spatio-temporal stability features and the cloud model. *CloudTrie* combines three BGP related data sources—RPKI, IRR, and RIB—and leverages a Trie tree for efficient storage and retrieval of prefix and ASN information. It uses a cloud model to assess route uncertainty degree in real time and incorporates an online update mechanism to adapt to the time-varying nature of BGP networks. The proposed method effectively handles the uncertainty in routing data and provides a more accurate and real-time AD. Experimental results demonstrate that *CloudTrie* achieves robust, adaptive, and highly accurate AD. Notably, *CloudTrie* provides uncertainty quantification for each generated alert. Our *CloudTrie* can be deployed by ISPs as monitoring instruments that link with BGP collectors, enabling network operators to more intelligently and quickly identify potential abnormal behaviors. We believe *CloudTrie* lays the foundation for more accurate and real-time AD in the global BGP network.

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